Summary Handout

Lecture 2

Psychology is the scientific study of the mind. There are various domains of psychology that attempt to understand the mind and apply this understanding to daily living. Although both understanding and application are relevant to our daily living, the knowledge accumulated in psychology is not based on gut feelings, personal experiences, speculations, or traditional beliefs. Thus, psychological studies are not eminence-based—based on the knowledge or opinion of well-respected psychologists. Rather, psychological studies are evidenced-based—based on empirical research evidence.

Questions have been raised about why psychological studies require empirical research evidence. A class activity helps illustrate how factors commonly viewed by the public as being associated with happiness are not necessarily well-supported by research. This implies that our common senses/intuitions or even public consensus are not always accurate. The following three key benefits are promised by research studies to help ascertain the validity of the research findings.

Key Benefits of Empirical Research Studies

1) Clarity and precision of study variable(s)

   In psychological research, the psychological constructs (i.e., concepts) of interest have to be clearly defined. The definition is based on an extensive review of past relevant literature. An example can be viewed in the case of "happiness" previously discussed in Activity 1. The reason why the class identified different factors contributing to happiness might result from their different interpretations of the construct "happiness."

   A review of past psychological literature helps identify two definitions of happiness. These are from 1) the subjective (hedonic) well-being perspective which defines happiness as positive feelings and evaluations of ones' life conditions, and 2) the psychological (eudaimonic) well-being perspective which defines happiness based on the degree to which ones live their lives based on their virtues and inherent potentials. Therefore, with different definitions of happiness, different factors are likely to be related to the construct.

2) Relatively low errors due to a systematic research procedure

   There are various steps in conducting research studies which help reduce the risks of errors. These steps start with a literature review which is an extensive review of existing information. Based on this information, researchers will set up a research question, for instance, what make kindergarten students happy. Then, they will make a certain proposition or formulate a hypothesis based on the literature review. For instance, they might hypothesize that receiving a candy will bring happiness to the students. Afterward, they will go out and collect data in the real world to examine whether the data support or reject their hypothesis. To clarify this, rigorous data analysis will be conducted using proper statistical procedures. By following these steps, together with careful selections of the definition of the psychological construct, proper research participants, and research tools, to name some, psychologists can become certain that the knowledge obtained through an empirical research study is relatively error-free and trustworthy.

3) Reviews by other researchers

In addition to the steps previously mentioned, the circulation and publication of worthwhile psychological knowledge are generally done in peer-reviewed psychological journals. This means that the publication will be reviewed and then approved by psychologists with in-dept knowledge in the domain. This helps ensure that the studies published are conceptually valid and methodologically sound.

Despite the benefits promised by research studies, there remain various areas in psychology that requires further investigation. This is because human mind is so complex that no single research study would be capable of illustrating its full complexity. Thus, in and of themselves, research studies are yet to be perfect. However, these various research processes still help maximize the validity of the research outcomes obtained.

Sample Research Processes

There are various processes involved in research investigations. These start with psychologists raising research questions. Then, based on a literature review, psychologists refine the questions, narrow them down, define study variable(s), and set up relevant research hypotheses. For example, if a psychologist aims to investigate what bring about happiness among kindergarteners, he will first engage in relevant literature review. Then, he will have to choose whether he would define happiness in terms of subjective or psychological well-being.

Drawing from his knowledge in developmental psychology, he is likely to decide that, for younger children whose self-concept and self-knowledge remain relatively limited, the former definition of happiness fits better. Hence, he may decide to do a survey of what types of objects (e.g., toys, food) lead to positive feelings in these young individuals. Considering their limited reading and writing capacities, rather than asking the kindergarteners to fill in a paper-and-pencil questionnaire, the psychologist may simply ask them to select from a list of three different smiley faces and point to the one that best reflects their positive feeling when seeing each object. This research tool is considered to be appropriate to the levels of the participants' development.

Upon obtaining the data, the psychologist will select and conduct proper data analysis. All in all, these processes need to be done within the ethical standards. One of the ethical considerations for this study is that the participants are very young, being kindergarteners, and thus they are unlikely to be capable of making a sound decision by themselves regarding whether they should participate in the study. Hence, parental permission for their study participation needs to be obtained in advance.

Research Designs in Psychology

Given the complexity of the human minds, various research designs have been proposed. Some commonly used designs are outlined below:

1) A case study: This study design is used with the objective to obtain a detailed, generally longitudinal, portrait of one or a few participants on a study variable, including some new/rare disorders or treatment innovations. Interviews, observations, and record checking are often used for data collection. An example is a study of Ted's rare illness or his aggression across lifespan.

The case study design draws its strengths from its provision of significant details. With its rich information, changes in the variable(s) of interest can be chronicled. Moreover, the design also helps provide substantial description of a rarely existing phenomenon.

Still, despite the aforementioned strengths regarding its intensive detailed focus, this design requires significant resources (e.g., time, budget). Additionally, due to the use of a small number of participants who might have potentially unique characteristics, the generalizability of the findings might be limited.

2) An experimental design: This study design is used with the objective to infer causes and effects by examining how the manipulated change(s) in a variable (i.e., an independent variable) impact another (i.e., a dependent variable). The design involves data collection from a larger pool of participants than in the case study design. Statistical analyses are conducted on the data obtained. An example is a study of how the aggression

levels of undergraduate students change (i.e., a dependent variable) when they get exposed to different levels of noises (i.e., an independent variable).

The experimental research design draws its strengths from its provision of a clear cause-and-effect inference. However, there are ethical concerns in some study manipulations—some manipulations could pose negative impacts on the participants. An example is from the Milgram's study of obedience (1961), where participants' obedience to authority (i.e., an independent variable) was investigated based on the manipulation of relevant factors (e.g., age and credibility of the authorities, the presence of protesters) (i.e., dependent variables). Whereas the study helped identify how these factors impact obedience, the participants, whose obedience was examined in terms of to what extent they would obey an instruction to administer potentially fatal electric shocks to others, could be adversely impacted psychologically. This poses an ethical limitation for the manipulation.

3) A correlational design: This study design is used with the objective to examine associations between two or more psychological variables. The outcomes could determine if changes in one variable are related to changes in another. Additionally, the degrees of the relationships (e.g., small, medium, or large) can be obtained, together with the directions of the associations (e.g., a positive correlation: when one variable increases, another does so; a negative correlation: when one variable increases, another decreases). An example is a study of how toddlers' aggression levels change in relation to the toddlers' exposure to television violence.

The use of the correlational research design helps overcome the ethical limitations of the experimental design. For example, due to the ethical reason, a researcher cannot purposely expose some toddlers to violent television programs in order to examine how aggressive they will become in comparison to those without such an exposure. However, using a correlational research design, the researcher can simply collect data on the toddlers' existing exposure to television violence and their corresponding aggression levels. Therefore, the correlational design draws its strengths from its applications to a broader array of psychological phenomena, when compared with an experimental research design.

However, the correlational outcomes obtained simply suggest bidirectional relationships between the study variables, *not* implying the causal one. Also, the outcomes obtained may result from a third external variable, rather than the relationship between the two variables. For example, the increasing temperature during summer season might be the third variable that accounts for the positive association between ice cream consumption and aggression levels (i.e., the association between an increase in the amount of ice cream consumption and higher levels of aggression can be explained by the increasing temperature).

Ethical Considerations

Although there are various research designs in psychological studies, all are still subjected to ethical considerations. Thus, the five ethical principles previously mentioned in Lecture 1 need to be considered. One of the research guidelines based on such principles is the need to obtain voluntary informed consent. Potential research participants have to be informed about the nature, scope, as well as potential risks and benefits of the study they are invited to participate. Then, they have to provide a voluntary consent before participating in the study—they cannot be forced or manipulated to do so. Still, after giving the consent, they can withdraw from the study at their free will. Moreover, after data are collected, researchers must maintain the confidentiality and anonymity of the participants' information. The outcomes of a study will be presented as a whole such that no direction/identification can be made to a particular participant. Lastly, all of the psychological research studies must be reviewed by an institutional ethics committee prior to being conducted.